

**COVID-19 Images**

Microscopic images of COVID-19 infected cells taken from the respiratory tract are published by scientists. <http://digit.in/mar21-41>

**Replacing Fossil Fuels**

Green hydrogen could make a practical replacement for fossil fuels. <http://digit.in/mar21-42>

Microscopy



Peer into the microcosmos

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A microscope (from Ancient Greek: mikrós, “small” and skopeîn, “to look” or “see”) is a laboratory instrument that is used to view objects that are too small to be seen with the naked eye. Microscopy is the science of using a microscopic lens to view small objects and structures.

HISTORY

What we know as the modern microscope can be first traced back to the Nimrud Lens. The Nimrud lens, also known as the Layard lens, is a 3000-year-old piece of rock crystal discovered in 1850 at the Assyrian palace of Nimrud in modern-day Iraq by Austen Henry Layard. It may have been a magnifying glass, a burning-glass used to start fires by focusing sunlight, or a piece of decorative inlay.

While the earliest known use of simple microscopes (magnifying

glasses) dates back to the common use of lenses in eyeglasses in the 13th century, there are Greek accounts of the optical properties of water-filled spheres (5th century BC), followed by several centuries of writings on optics. About 1620, the first known examples of compound microscopes appeared in Europe, which combined an objective lens near the specimen with an eyepiece to display a real image. While several arguments have been made over the years, the inventor remains elusive.

Galileo Galilei (often referred to as the compound microscope inventor) discovered that he could closely focus his telescope to view small objects and built his own improved version after seeing a compound microscope built by Drebbel exhibited in Rome in 1624. For the compound microscope Galileo submitted to the Accademia dei Lincei in 1625, Giovanni Faber coined the term microscope (Galileo had called it the “occholino” or “little eye”).

At this time, microscopes were far from ideal. When viewed through a microscope, objects didn’t have enough resolution, and there was visual aberration around the edges. That’s where J. J. Lister comes in. J. J. Lister

was intrigued by natural history and found that the microscopes available in the early 1800s lacked the precision required to show the structure of plant and animal cells in adequate detail. As a result, he set out to design and create superior achromatic lenses, mixing lenses of crown and flint glasses of different dispersion to cancel chromatic aberration, demonstrating that spherical aberration could be reduced by proper separation of the lens combinations, leading to the optical microscope’s perfection.

ABBE’S EQUATION

Ernst Karl Abbe was a German mathematician and physicist who met Carl Zeiss in 1866 and founded the ‘Zeiss Optical Works,’ which is now known as Zeiss. He was also a co-founder of Schott Glassworks in 1884. Abbe is often credited with coining the term numerical aperture. Abbe published his theory and formula for explaining the microscope’s diffraction limits in 1873. Abbe recognised that specimen images are made up of several overlapping, multi-intensity, diffraction-limited points (or Airy Discs).

Even with all of these considerations taken into account, the limits in a real microscope device were still very constrained due to the overall system’s complexity.

The idea for an imaging instrument that would image by exciting and collecting diffraction in the near field was conceived and developed by Edward Hutchinson Synge. His initial design was to use extreme nearly planar light from an arc under pressure behind a thin, opaque metal film with a narrow orifice of about 100 nm, which he introduced in 1928. The orifice had to stay within 100 nanometers of the surface, and data